

# Generative AI as Seniority-Biased Technological Change: Evidence from U.S. Résumé and Job Posting Data\*

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## Abstract

This paper studies whether generative AI (GenAI) constitutes *seniority-biased technological change*, disproportionately reducing demand for junior workers. We develop a conceptual framework in which GenAI adoption reduces junior labor demand through task displacement and labor-saving productivity gains. We test the framework’s mechanisms and implications using U.S. résumé data covering 65 million workers at more than 280,000 firms, allowing us to track firm-level employment by seniority. GenAI adoption is identified through text analysis that detects ‘GenAI integrator’ job postings, signaling active GenAI implementation by firms. Following adoption, junior employment declines in adopting firms relative to non-adopters, while senior employment trends remain largely unchanged. This decline is concentrated in occupations most exposed to GenAI and is driven primarily by slower hiring rather than increased separations. Moreover, in adopting firms, GenAI-exposed tasks become increasingly less likely to appear in junior task bundles.

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to senior employment. Before turning to this aggregate evidence, we first examine the framework’s underlying mechanisms. Using an LLM, we map a representative sample of our job postings to standardized O\*NET tasks and show that AI-exposed tasks become disproportionately less likely to appear in junior task bundles at adopting firms after 2023, while no comparable pattern emerges for seniors. This provides evidence consistent with the displacement channel. By contrast, tasks that were ex ante concentrated among seniors do not systematically migrate into junior task bundles, offering little support for the reallocation channel. We complement these tests with external task-level data from the Anthropic Economic Index (Appel et al., 2026), documenting that real-world GenAI use is more likely to take the form of automation for tasks characteristic of junior occupations, whereas usage in senior occupations is more often augmentative, consistent with the framework’s assumption that junior tasks are more susceptible to displacement.

We then turn to the firm-level employment response. We first compare junior and senior employment at adopting and non-adopting firms. From 2018 through 2022, junior employment evolves similarly in the two groups. Beginning in late 2022, however, junior employment at adopters diverges sharply downward relative to non-adopters, while senior employment shows no comparable break. To directly assess the seniority-biased effect, we estimate a triple-difference specification that compares junior versus senior employment within adopting firms relative to non-adopters. This design includes firm-by-time fixed effects, which absorb shocks or trajectories specific to a given firm in a given period, and industry-by-time-by-seniority fixed effects, which account for sector-level dynamics that differentially affect juniors and seniors. The coefficients are flat through 2022Q4 and then decline sharply from 2023Q1 onward, reaching roughly a 9 percent decline in junior employment relative to senior employment after six quarters.

We next examine which types of jobs drive our results, focusing on heterogeneity by occupational exposure to GenAI using the widely used exposure measure of Eloundou et al. (2024). Specifically, we estimate a triple-difference specification that compares employment changes in high- versus low-exposure occupations within adopting and non-adopting firms, separately for junior and senior workers. Through the end of 2022, the estimated coefficients for juniors and seniors follow nearly identical trajectories. Beginning in 2023Q1, however, the two series diverge sharply. Junior employment in high-exposure occupations declines by roughly 7 percent relative to low-exposure occupations over the

subsequent two years, while the corresponding senior series continues to rise. This pattern is consistent with GenAI substituting for exposed junior tasks while leaving demand for senior workers in those occupations largely unaffected.<sup>6</sup>

We complement these analyses with a staggered event study that traces employment dynamics around firms' GenAI adoption date, proxied by the first period in which the firm posts a GenAI integrator role. This design helps distinguish adoption effects from broader time-specific shocks by exploiting variation in the timing of adoption across firms. However, it is sensitive to measurement error in the adoption timing proxy—for instance, if firms begin using GenAI before, or only several periods after, posting for an integrator role. The estimates reveal no evidence of differential pre-trends: employment evolves similarly in adopting and non-adopting firms throughout the eight quarters preceding adoption. However, roughly two quarters after adoption, junior employment in adopting firms begins to decline, reaching an 8 percent reduction after eight quarters. The absence of significant pre-trends provides additional reassurance that these post-adoption declines are unlikely to be driven by confounding shocks.

We next address potential endogeneity concerns. Adopting and non-adopting firms differ systematically, which could bias our estimates; for example, if adopting firms are more responsive to other shocks affecting employment. One concern is that GenAI adopters may have been more sensitive to the 2022–2023 monetary tightening cycle. Several pieces of evidence suggest that this mechanism does not drive our findings: including industry-by-seniority-by-time fixed effects leaves the results essentially unchanged; the event-study estimates show no pre-adoption trends; and a formal test based on the high-frequency identified monetary-policy surprises of [Jarocinski and Karadi \(2020\)](#), following [Ottonello and Winberry \(2020\)](#), finds no evidence that adopter firms are more responsive to monetary shocks. This result is also consistent with the literature documenting that larger firms—which are more likely to adopt GenAI—tend to be less sensitive to monetary policy ([Chodorow-Reich, 2014](#); [Gertler and Gilchrist, 1994](#)).

We also consider three interrelated concerns arising from the post-pandemic labor market: the hiring boom and subsequent correction in the technology sector, the expan-

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<sup>6</sup>Supplemental Appendix [A.3](#) reports a complementary heterogeneity exercise by juniors' educational background. Measuring school quality by the prestige of juniors' alma mater, we find a U-shaped pattern: juniors from mid-tier universities experienced the largest relative declines, while those from the most and least prestigious institutions saw smaller reductions.

## A.3 Heterogeneity by Educational Background

This appendix examines whether the decline in junior employment varies by workers' educational background, as referenced in Section 5.2.3. We first describe the construction of the school-quality variable, then present the heterogeneity results.

### A.3.1 Construction of the School-Quality Variable

To capture educational background, we construct a position-level measure of school quality. For each position, we assign the institution where the worker most recently studied, provided that the education ended no later than one year after the job start. If no such record exists, we use the first recorded institution, provided it began before the job start date. Each institution is then assigned a GPT-4-based quality rating on a five-point scale, produced using OpenAI's GPT-4o-mini model (see Supplemental Appendix A.11.2 for the full prompt). A score of 1 corresponds to Ivy League and other globally elite institutions; 2 to highly respected international universities; 3 to strong national or regional schools; 4 to lower-tier but standard institutions; and 5 to minimally selective or non-accredited institutions.

### A.3.2 Results

We re-estimate Equation 34 for junior employment separately within each of the five school-quality tiers. For clarity of presentation, we use a single post-treatment indicator for quarters after 2023Q1 instead of a full set of time dummies.

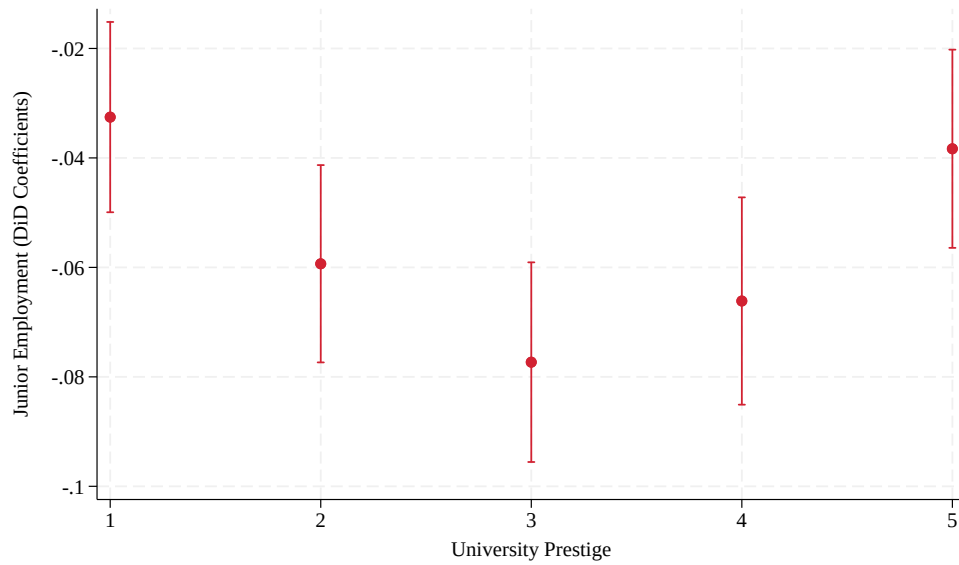
The results, shown in Figure A.12, reveal a U-shaped pattern. Juniors from tier-3 universities experienced the steepest relative declines in employment, while those from tiers 2 and 4 also saw significant reductions. Juniors from tier-1 and tier-5 universities experienced the smallest decline.<sup>33</sup> Figure A.13 reports the time-series version of these estimates, with a full set of time dummies in place of the single post-2023 indicator.

To help interpret this pattern, Figures A.14 and A.15 show average predicted salaries<sup>34</sup>

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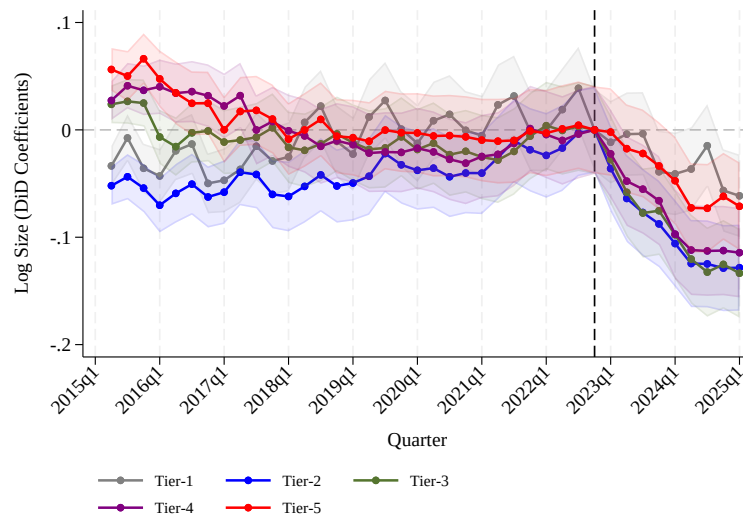
<sup>33</sup>In the analysis, we exclude positions where the worker's most recent educational entry is reported as a high school (approximately 5 percent of positions). Including these positions, based on high school quality, has virtually no effect on the results.

<sup>34</sup>Actual position-level salaries are unavailable. However, Revelio Labs provides a predicted salary



**Figure A.12: Heterogeneity by School Quality**

*Notes:* Each bar reports the estimated coefficient from re-estimating Equation 34 for junior employment separately within each of the five school-quality tiers. The specification replaces time dummies with a single post-treatment indicator for quarters after 2023Q1. Estimation begins in 2018 to exclude early-sample trends. Standard errors are clustered at the firm level.



**Figure A.13: DiD by Educational Background—Time Series (Juniors)**

*Notes:* This figure plots  $\beta_j$  from Equation 34, estimated separately for each of the five school-quality tiers. Standard errors are clustered at the firm level.

variable imputed using position-specific information. For more details, see: <https://www.data-dictionary.reveliolabs.com/methodology.html#salary>.